

Course Code : ECT 456-3

QTSR/RW – 24 / 1808

**Eighth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

ANTENNA THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) Assume suitable data wherever necessary.
- (2) All questions are compulsory.

1.
 - (a) Explain the special features of various types of Horn antennas. 5(CO5)
 - (b) Discuss any three parameters of antenna using neat diagrams. 5(CO1)
2.
 - (a) Discuss the effects of dielectric constant while designing microstrip patch antenna. 5(CO4)
 - (b) Analyze the constructional differences in Yagi-Uda and log periodic antenna with respect to changes in bandwidth and gain. 5(CO2)
3.
 - (a) Explain advantages and disadvantages of corner and flat sheet reflectors. Comment on directivity in each case. 5(CO1)
 - (b) Apply Huygen's principle in designing an aperture antenna. 5(CO3)
4.
 - (a) For end fire array consisting of several half wave length long isotropic radiation, to have directive gain of 30. Find the array length and width of the major lobe (i.e. beam width between nulls). What will be these values for broadside array. 5(CO4,5)
 - (b) Calculate the directivity of a linear endfire and broadside array with uniform array of 10 elements with $\lambda/4$ spacing between them. 5(CO4,5)

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Contd.

5. (a) Explain advantages and disadvantages of smart antennas with respect to Cellular Radio Systems. 5(CO3)
- (b) Discuss cell splitting and Sectorized Systems with respect to smart antennas. 5(CO3)
6. (a) Explain signal propagation in smart antennas. 5(CO3)
- (b) Give an overview of Mobile Ad hoc NETWORKs (MANETs) with respect to smart antennas. 5(CO3)

